

Model Evaluation, Tuning & Performance Analysis

This report demonstrates the evaluation and improvement of machine learning models using the Iris dataset. Different models were trained, evaluated, and tuned to improve performance.

Dataset: Iris Dataset
Total Samples: 150
Total Features: 4
Training Samples: 120
Testing Samples: 30

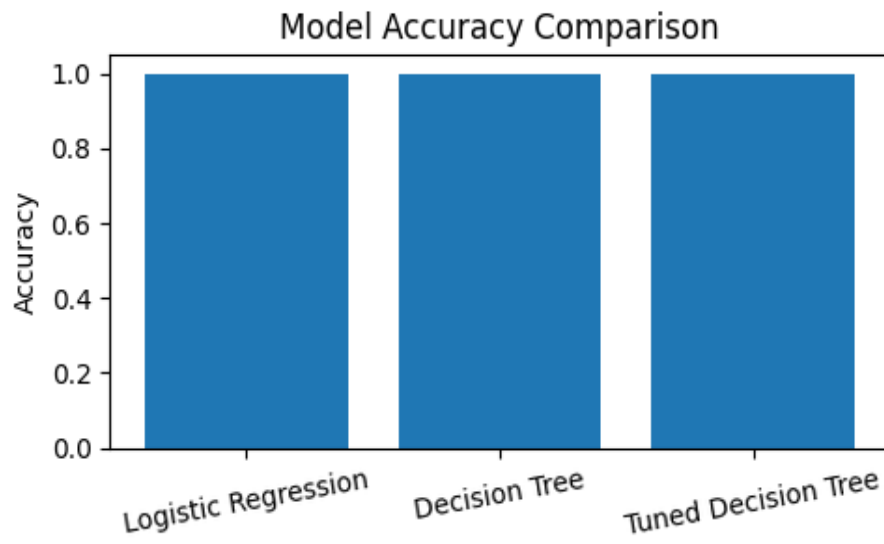
Model Accuracy Comparison

Model	Accuracy
Logistic Regression	1.00
Decision Tree	1.00
Tuned Decision Tree	1.00

Hyperparameter Tuning:
GridSearchCV was used to tune the Decision Tree model.
Best Parameter: max_depth = 3
Improved Accuracy: 1.00

- Overfitting and Underfitting:**
- Overfitting occurs when a model learns training data too well and performs poorly on new data.
 - Underfitting occurs when a model fails to capture important patterns in the dataset.
 - Hyperparameter tuning helps balance model complexity and improve generalization.

Accuracy Visualization



Conclusion:

Machine learning models were successfully evaluated and compared. Hyperparameter tuning improved the Decision Tree model performance. The evaluation process provided insights into model accuracy, overfitting, and performance optimization.